

Design, Modeling, and Control of a Coaxial Drone

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I. INTRODUCTION

Abstract—Various quadrotor drones have been developed in recent years, mainly focusing on either improving maximum thrust per platform area or flight maneuverability. Evidently, achieving both advantages simultaneously is a challenging task, since they call for opposing rotor requirements. Specifically, improving the drone's maximum thrust per platform area mainly requires reducing the number of rotors to make way for larger and more powerful rotors. While this can be an effective method to increase overall thrust, improving flight maneuverability requires a greater number of rotors to generate larger rotating torques or to increase the thrust vectoring capability. To address this challenge, we design a novel coaxial drone with two contra-rotating rotors for high thrust efficiency while enabling independent dual-axis rotor rotation to maintain maneuverability along the roll and pitch axes. The thrust vectoring capability is provided by two dedicated servomotors connected vertically in series with the coaxial propellers to produce a compact and elongated fuselage frame. A nonlinear flight model in six degrees of freedom is developed for the underactuated system, incorporating four control inputs from the two propellers and servos, respectively. Consequently, a nonlinear control allocation approach is proposed such that the drone can produce a desired control force and yaw torque to stabilize the drone's position and yaw angle. For the uncontrolled roll and pitch dynamics, a damping component is added such that the roll and pitch angular velocities can also be stabilized. Both numerical simulations and real experiments are conducted to validate the design of the drone and the effectiveness of the proposed control strategy.

Index Terms—Coaxial drone, control allocation, drone design, dynamics modeling, flight control, thrust efficiency.

WITH the recent technological advancements in aerial vehicles' power supply, communication, and onboard computation, the use of quadrotors in various engineering applications has become increasingly popular [1]. In particular, quadrotors can provide increased safety and reduce the cost of operations in high-risk tasks or environments, such as disaster management [2], search and rescue [3], bridge and building inspection [4].

The conventional quadrotor design adopts a plus or cross configuration with two pairs of counter-rotating propellers connected to a central body. Owing to its mechanical simplicity, quadrotors enjoy a high stability in hover and a high maneuverability in controlled movements involving roll and pitch motions. However, the maximum thrust per platform area, i.e., thrust efficiency, for quadrotors is generally limited due to the presence of four propellers, whose rotor size is proportional to the central body. Note that the maximum thrust per platform area is an important index for a drone's flight duration and power consumption [5].

Subsequently, this drives the motivation to design a new drone, which can achieve higher thrust per platform area while still preserving the same maneuverable degrees of freedom (DoF) as quadrotors. First, for the thrust efficiency, since the power needed to hover is proportional to the inverse of the rotor's radius, a single large rotor will energetically outperform several smaller rotors within the same boundary [5]. Motivated by this fact, a triangular quadrotor configuration is proposed in [5] by using a single large rotor for lift and three small rotors for control. Also, a bi-copter drone is designed in [6], which has similar levels of versatility and improves the drone's compactness or efficiency at the same time. Second, to further increase drones' maneuverable DoF, tilting rotors are employed for the drone design [7], [8], [9], [10]. For example, an overactuated quadrotor is developed in [8] based on four tilting rotors. Since coaxial contra-rotating rotors can provide a compact and anti-torque solution [11], a coaxial drone is one of the promising options to simultaneously satisfy these two requirements, i.e., higher thrust per platform area and the same maneuverable DoF as quadrotors [12]. The effect of coaxial propellers' configuration on propulsion efficiency is discussed in [13], which is defined as the multiplication of the propeller's efficiency, the motor's efficiency, and the electronic speed controller's efficiency.

Several coaxial drones have been developed in [11], [12], [14], [15], [16], one of which is the NASA's Ingenuity Mars helicopter, which successfully completed its first flight test in early 2021. The Ingenuity helicopter has contra-rotating coaxial

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propellers connected to a lightweight body frame. With a high thrust efficiency, the coaxial design of Ingenuity allows it to perform basic takeoff and landing maneuvers under the thin atmosphere and restrictive energy equipment in Mars¹. Unlike the coaxial quadrotor drone in [16], other coaxial drones designed in [11], [12], [14], [15] contain coaxial propellers, whose roll and pitch motions are controlled by their designed mechanical swashplates. Similar to the coaxial drones in [11], [12], [14], [15], the roll and pitch motions of the Ingenuity helicopter are independently controlled by two cyclic swashplates¹, each of which is driven by three small servomotors. However, the use of two swashplates with six servomotors involves more intricate moving parts that increase the overall mechanical complexity of the platform. As a result, with an increased control complexity of the drone, the probability of failure is also escalated.

Inspired by existing coaxial drones [11], [12], [14], [15], and the tilting mechanism [6], [7], [8], [9], [10] employed in drone design, this article presents a compact drone with contra-rotating coaxial propellers that is capable of achieving independent bi-axial rotation about two orthogonal axes. This design preserves the four maneuvering DoF with a higher thrust per platform area than that of a conventional quadrotor. Unlike the cyclic swashplate mechanism used in other coaxial drones, only two servomotors are directly employed to provide thrust vectoring capability in producing the roll and pitch motions. This enables the platform to be controlled in a more intuitive and mechanically simpler way. Another potential advantage of using servomotors for the control of roll and pitch motions is their larger outputted angle range than that of swashplates. Based on the integration of the coaxial propellers with the two servomotors connected in series, a detailed six-DoF model involving the drone's position and attitude dynamics is developed. Then, to design an efficient controller for the underactuated system based on the four control inputs given by the two propellers and two servos, respectively, a nonlinear control allocation approach is proposed, under which the drone can produce desired control force and yaw torque to stabilize the drone's position and yaw angle. Finally, we validate our established model and proposed control strategy through both numerical simulations and real-flight experiments.

The rest of this article is organized as follows. Section II introduces the overall design of the coaxial drone. Section III establishes the attitude and position dynamics for the coaxial drone. The control design is provided in Section IV. Sections V and VI present the simulation results and experimental results, respectively. Finally, Section VII concludes this article.

II. DESIGN OF THE COAXIAL DRONE

In this section, we introduce the design of the coaxial drone. First, we present the drone's overall mechanical structure and then introduce different rigid bodies in the coaxial drone and their respective coordinate frames. Finally, we discuss the connection of the coaxial propeller system, servomotors, and the drone's fuselage.

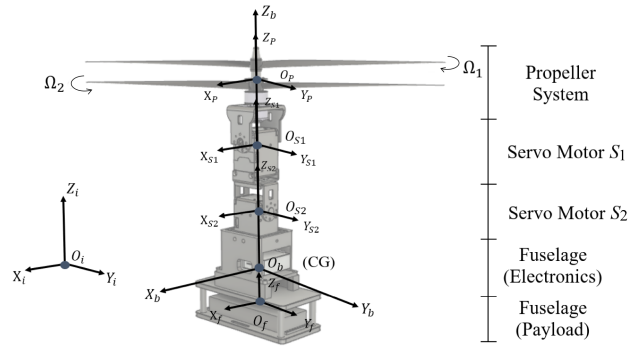


Fig. 1. Overall structure of the coaxial drone.

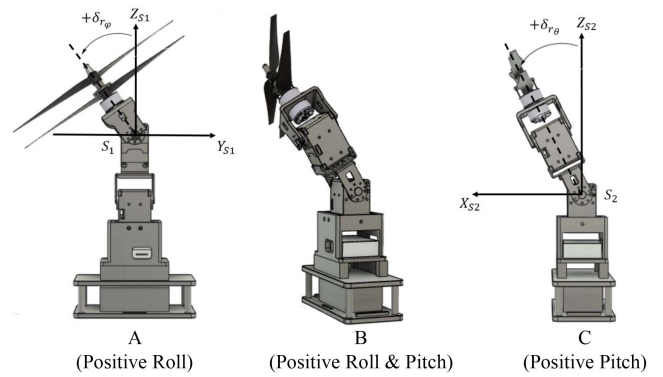


Fig. 2. Two servomotors' rotating angles.

A. Overall Mechanical Structure

The coaxial drone consists of one propeller system, two servomotors and one lower fuselage section. The coaxial propeller system consists of two counter-rotating rotors powered by a pair of brushless motors that generate necessary thrust for flight. The two servomotors produce localized roll and pitch rotations of the propeller system for thrust vectoring. A series of connecting mounts also provide structural support for the coaxial drone. The drone's fuselage section houses the flight controller, battery, and other electrical components. The components located in the drone's fuselage have a relatively larger mass compared to the servomotors and the propeller, resulting in the center of gravity (CG) of the entire coaxial drone to be located in the fuselage section. We assume that the CG of the entire drone is vertically aligned with the CG of the drone's fuselage. A representation of the coaxial drone and its localized rotation angles are shown in Figs. 1 and 2. In addition, since the airflow produced by the propeller system is towards the direction from the propeller to the fuselage, the fuselage part is designed to be as vertical as possible, ensuring minimum obstruction to the airflow.

B. Rigid Bodies and Coordinate Frames

As shown in Fig. 1, there are four rigid bodies assembled together to form the coaxial drone. We can introduce coordinate frames for each of them to describe their relative positions in space. The first rigid body consists of the propeller system and its rotation mount. The corresponding frame is represented

¹[Online]. Available: <https://mars.nasa.gov/technology/helicopter/status/336/2800-rpm-spin-a-success-but-flight-14-delayed-to-post-conjunction/>

by $\sum_p : O_p - X_p Y_p Z_p$ whose O_p is fixed to the center of the contra-rotating rotors. The second rigid body consists of the first servomotor S_1 and the rotation mounts attached to its bottom surface. The corresponding frame is represented by $\sum_{S1} : O_{S1} - X_{S1} Y_{S1} Z_{S1}$ with O_{S1} fixed to the center of rotation axis of S_1 . In addition, S_1 controls the roll angle δ_{r_ϕ} of the first rigid body O_p , where a counterclockwise rotation along X_{S1} produces a positive roll (i.e., following the right-hand screw rule). The third rigid body consists of the second servomotor S_2 and its base mount, which is connected to the drone's fuselage. It has a frame of reference $\sum_{S2} : O_{S2} - X_{S2} Y_{S2} Z_{S2}$ with O_{S2} fixed to the center of rotation axis of S_2 . Also, S_2 controls the pitch angle δ_{r_θ} of the second rigid body O_{S1} where a counterclockwise rotation along Y_{S2} produces a positive pitch. The fourth rigid body refers to the drone's fuselage and its respecting components. It has a body frame $\sum_f : O_f - X_f Y_f Z_f$ with O_f being fixed to the CG of the rigid body.

In the follow-up sections of this article, two frames are selected as the main reference frames for the modeling of the drone. The first is the inertial coordinate frame $\sum_i : O_i - X_i Y_i Z_i$ fixed to the ground of Earth. The second is the body frame $\sum_b : O_b - X_b Y_b Z_b$ whose origin O_b is rigidly attached to the CG of the coaxial drone (the total mass of the drone is taken to act at this point).

C. Connection of Coaxial Propeller, Servomotors, and Fuselage

The coaxial drone has a simple assembly, with its mechanical and electrical components connected in a vertical arrangement. The coaxial propeller system serves as the primary source of thrust for the drone along a common axis. A carefully designed rotation mount provides sufficient clearance for the unobstructed rotation of the propeller by $\delta_{r_\phi} \in [-90^\circ, 90^\circ]$, as shown in Fig. 2. A stronger rotation mount is used for the secondary servomotor S_2 to support the combined weight of the propeller system and S_1 . Similarly, Fig. 2 also shows that a rotation clearance of $\delta_{r_\theta} \in [-90^\circ, 90^\circ]$ can be achieved by the subsequent mount in the system, which indicates that the outputted angle range of servomotors is larger than that of swashplates. This dual arrangement of the servomotors allows the drone to perform forward-backward and lateral maneuvers in flight by rotating the direction of the generated thrust. The usage of the angles δ_{r_ϕ} and δ_{r_θ} to maintain flight stability of the drone is explored in the follow-up sections of this article.

Another consideration for the assembly of the drone is the design of the connecting base mount (see the component between O_{S2} and O_b in Fig. 1) attached between the end of S_2 and the drone's fuselage. The dimensions of the base mount are designed to allow for a secure connection for ease of assembly and effective transfer of forces. The drone's fuselage consists of two housing frames that contain the flight controller and the battery pack, respectively. All remaining electrical cables and connecting components are stored within the drone's fuselage as well.

D. Comparison With the Existing Design of Coaxial Drones

The main difference between the coaxial drone introduced in this article and other birotor coaxial drones developed in

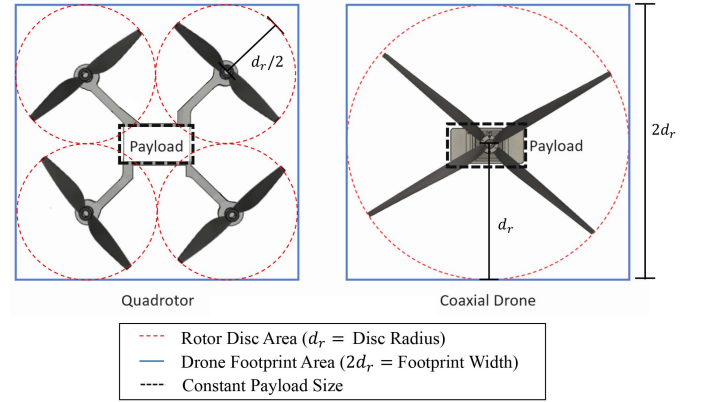


Fig. 3. Comparison between a quadrotor and the coaxial drone.

[11], [15] lies in the mechanism designed to control the thrust vectoring of the propeller system. In [11], [15], the proposed coaxial drones utilize a swashplate platform to control their direction of flight, in which servomotors attached to the platform change the pitch of the propeller blades at each rotation cycle and then generate thrust components parallel to the desired direction. Through this mechanism, thrust vectoring is indirectly controlled by the servomotors. In [12, Section 2], two servomotors in a parallel mechanism are used to control the tilt of the swashplate. In [14], a bioinspired reaction wheel mechanism is employed for the design of a coaxial helicopter. In this article, the proposed coaxial drone design has two servomotors connected directly to the propeller system through the use of individual rotation mounts, under which the parallel coaxial propellers do not need to tilt with respect to the rotors' brushless motor to achieve thrust vectoring. The main advantage of this design is that it is more intuitive and mechanically simpler in maneuvering the drone toward the desired flight direction. The increase in drone height due to the vertical arrangement of the servomotors is compensated by the compact design of the drone fuselage. The fuselage is assembled as modular sections, which also reduces the drone's weight as opposed to the bulky cylindrical housing, as shown in [12] and [15]. The usage of swashplates involves multiple moving parts actuated by servomotors and hence has a higher probability of failure. For example, the Ingenuity helicopter has two swashplates for independent pitch control of each rotor and is required to perform an automated check on all six servos before each flight. Another difference of using servomotors to generate the drone's roll and pitch motions is their larger outputted angle range than that of swashplates, resulting in a potential advantage and improved degree of control, as shown in Fig. 2.

E. Comparison of Thrust Per Platform Area With Quadrotors

In this section, we compare the maximum thrust per platform area between a quadrotor and the coaxial drone with similar payload capabilities, as shown in Fig. 3. To have a fair comparison, we consider that both of the quadrotor and the coaxial drone are designed with the same footprint area (characterized by a square), the same size of payload, the same overall weight,

and rotational angular speed. Fig. 3 shows that the radius of the quadrotor's propeller is $d_r/2$, while that of the coaxial drone is d_r . According to the aerodynamic actuator disk theory [17], the magnitude of thrust generated by one rotor of the quadrotor can be calculated by

$$F_{\text{propeller1}} = \rho C_T \pi (d_r/2)^2 (\omega d_r/2)^2 = \rho C_T \pi \omega^2 d_r^4 / 16 \quad (1)$$

where ρ is the density of air, C_T is the lift coefficient, and ω is the angular velocity of the propeller. Then, the total thrust produced by a quadrotor is $F_{\text{quadrotor}} = 4F_{\text{propeller1}} = \rho C_T \pi \omega^2 d_r^4 / 4$. By using similar calculations, the ideal thrust produced by the coaxial drone is $F_{\text{coaxial}} = 2F_{\text{propeller2}} = 2\rho C_T \pi \omega^2 d_r^4$, where $F_{\text{propeller2}}$ represents the thrust produced by each propeller in the coaxial drone and the interaction effect between the two propellers is neglected. In practice, the thrust loss caused by the interaction effect is usually between 25% and 35%² (it is claimed in [18] that the thrust loss is about 23%). Thus, although we take into account the maximum thrust loss caused by the interaction effect between the propellers [19], the produced thrust $0.65F_{\text{coaxial}} = 1.3\rho C_T \pi \omega^2 d_r^4$ of the coaxial drone is still larger than that of the quadrotor. We will compare the produced thrust between the coaxial drone and quadrotors in the experimental section of this article.

Remark 1: In Fig. 3, the footprint of the drone is characterized by a square. If the platform footprint is characterized by a circle instead of a square, then the footprint of the coaxial drone will decrease to πd_r^2 , but the footprint of the quadrotor will increase to $\pi(\frac{1+\sqrt{2}}{2}d_r)^2$ according to the geometric relation. This will further make the coaxial drone's maximum thrust per platform area even larger, and the quadrotor's maximum thrust per platform area even smaller, as compared to the case of using square footprint.

Remark 2: A drone's real flight-time is related to many factors, including the drone's equipped battery, mass, size, and mechanical structure. To have a fair comparison of flight-time for different drones, we need to keep all these factors the same, which is difficult since different types of drones have different mass/sensors/components/actuators/batteries. As an alternative, the index of power consumption is used in [6] to conduct the comparison of power consumption among common multirotor drones, which is based on the momentum theory and the condition that all the drones share the same takeoff weight and platform footprint. From [6, Fig. 4], one can see that the power consumption of the coaxial drone is 0.707 times smaller than that of the quadrotor.

III. MODELING OF THE COAXIAL DRONE

To establish the coaxial drone's position and attitude dynamics, which characterize the relationship between the four control inputs ($\Omega_1, \Omega_2, \delta_{r_\phi}$ and δ_{r_θ}) and the resulting accelerations with six DoF, we refer to the coordinate frames introduced in Section II. As shown in Fig. 1, the body frame $\sum_b : O_b - X_b Y_b Z_b$ whose origin O_b is rigidly attached to the CG of the coaxial

drone, X_b -axis points toward the drone's forward direction, Y_b -axis points toward the drone's right direction, and Z_b -axis follows the right-hand rule which points towards the propeller if $\delta_{r_\phi} = \delta_{r_\theta} = 0$. The attitude and position dynamics in this modeling are described with respect to the coordinate frame $\sum_i$.

A. Attitude Kinematics and Dynamics

The attitude of the drone is described by three Euler angles $\mathbf{a} = [\phi, \theta, \psi]^\top \in \mathbb{R}^3$, which characterize the rotation from $\sum_b$ to $\sum_i$. According to mechanics of rigid bodies, the angular velocity $\dot{\mathbf{a}}$ of the drone in $\sum_i$ can be transformed from $\sum_b$ to $\sum_i$ using the equation

$$\dot{\mathbf{a}} = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & S_\phi T_\theta & C_\phi T_\theta \\ 0 & C_\phi & -S_\phi \\ 0 & S_\phi \sec \theta & C_\phi \sec \theta \end{bmatrix} \begin{bmatrix} \omega_{xb} \\ \omega_{yb} \\ \omega_{zb} \end{bmatrix} = \bar{\mathbf{R}} \boldsymbol{\omega}_b \quad (2)$$

where $\boldsymbol{\omega}_b = [\omega_{xb}, \omega_{yb}, \omega_{zb}]^\top \in \mathbb{R}^3$ represents the angular velocity vector consisting of roll, pitch, and yaw change rates of the drone in $\sum_b$, and $\bar{\mathbf{R}}$ is a three-by-three transformation matrix, and for notation conciseness $C : \cos$, $S : \sin$, $T : \tan$. Then, from (2), one has

$$\boldsymbol{\omega}_b = \mathbf{R} \dot{\mathbf{a}} = \bar{\mathbf{R}}^{-1} \dot{\mathbf{a}} = \begin{bmatrix} 1 & 0 & -S_\theta \\ 0 & C_\phi & S_\phi C_\theta \\ 0 & -S_\phi & C_\phi C_\theta \end{bmatrix} \dot{\mathbf{a}}. \quad (3)$$

According to the conservation theorem of angular momentum, one has

$$\left(\frac{d\mathbf{H}}{dt} \right)_i = \left(\frac{d\mathbf{H}}{dt} \right)_b + \boldsymbol{\omega}_b \times (\mathbf{H})_b \quad (4)$$

where $\mathbf{H} \in \mathbb{R}^3$ is the angular momentum of the drone, $(\frac{d\mathbf{H}}{dt})_i$ and $(\frac{d\mathbf{H}}{dt})_b$ are the time-derivatives of the angular momentum described in $\sum_i$ and $\sum_b$, respectively, $\times$ represents the operation of cross product, and $(\mathbf{H})_b$ represents the angular momentum in $\sum_b$. According to the definition of angular momentum, the changing rate of the drone's angular momentum in the inertial frame, $(\frac{d\mathbf{H}}{dt})_i = \mathbf{I} \ddot{\mathbf{a}}$, is equal to the sum of the torque generated from the rotation of the drone's body frame, $\boldsymbol{\omega}_b \times (\mathbf{H})_b = \boldsymbol{\omega}_b \times \mathbf{I} \boldsymbol{\omega}_b$, and the changing rate of the drone's angular momentum in its body frame $(\frac{d\mathbf{H}}{dt})_b = \mathbf{T}_t^b + \mathbf{T}_r^b$, where $\mathbf{I} \in \mathbb{R}^{3 \times 3}$ is the inertia of the drone which is assumed as a constant matrix, $\mathbf{T}_t^b \in \mathbb{R}^3$ and $\mathbf{T}_r^b \in \mathbb{R}^3$ are the applied torques to the CG described in $\sum_b$, which are produced by the coaxial propellers' thrust and rotation, respectively. Then, according to the coaxial drone's mechanical structure in Fig. 1, one has

$$\mathbf{T}_t^b = \mathbf{r}_{F_t} \times \mathbf{F}_t^b \quad (5)$$

where $\mathbf{F}_t^b \in \mathbb{R}^3$ is the thrust produced by the propeller system in $\sum_b$ whose specific form will be introduced later and $\mathbf{r}_{F_t}$ is the position vector describing the point of thrust with respect to the reference frame $\sum_b$. It can also be described as the vertical distance of O_p with respect to $\sum_b$ which can be derived from a series of transformation from $\sum_{S1}$ and $\sum_{S2}$ to $\sum_b$ as shown

²[Online]. Available: <https://www.tytorobotics.com/blogs/general-knowledge/coaxial-setup-and-testing>

in the following:

$$\mathbf{O}_p^{S_1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \delta_{r_\phi} & -\sin \delta_{r_\phi} \\ 0 & \sin \delta_{r_\phi} & \cos \delta_{r_\phi} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ d_1 \end{bmatrix} \quad (6)$$

$$\mathbf{O}_p^{S_2} = \begin{bmatrix} \cos \delta_{r_\theta} & 0 & \sin \delta_{r_\theta} \\ 0 & 1 & 0 \\ -\sin \delta_{r_\theta} & 0 & \cos \delta_{r_\theta} \end{bmatrix} \left[\begin{bmatrix} 0 \\ 0 \\ d_2 \end{bmatrix} + \mathbf{O}_p^{S_1} \right] \quad (7)$$

$$\mathbf{r}_{F_t} = \begin{bmatrix} 0 \\ 0 \\ d_3 \end{bmatrix} + \mathbf{O}_p^{S_2} = \begin{bmatrix} (d_1 \cos \delta_{r_\phi} + d_2) \sin \delta_{r_\theta} \\ -d_1 \sin \delta_{r_\phi} \\ (d_1 \cos \delta_{r_\phi} + d_2) \cos \delta_{r_\theta} + d_3 \end{bmatrix} \quad (8)$$

where d_1 is the absolute distance from O_p to S_1 , d_2 is the absolute distance from S_1 to S_2 , and d_3 is the absolute distance from S_2 to O_b . Note that (7) and (8) represent the coordinate transformations required to account for the roll and pitch rotations along the X_{S_1} - and Y_{S_2} -axis, respectively. A final translation of d_3 along the direction Z_b in (8) produces the position vector $\mathbf{r}_{F_t}$ with respect to the reference frame $\sum_b$. Also, the control torque induced by the rotation of the contra-rotating propellers can be described as

$$\mathbf{T}_r^b = \sigma(\gamma_1 \Omega_1^2 - \gamma_2 \Omega_2^2) \begin{bmatrix} S_{\delta_{r_\theta}} C_{\delta_{r_\phi}} \\ -S_{\delta_{r_\phi}} \\ C_{\delta_{r_\phi}} C_{\delta_{r_\theta}} \end{bmatrix} \quad (9)$$

where $\gamma_1 \in \mathbb{R}^+$ and $\gamma_2 \in \mathbb{R}^+$ are the lumped moment coefficients of the upper and lower propellers, respectively, and $0 < \sigma < 1$ is the aerodynamic efficiency factor due to the existence of airflow interactions, and $\Omega_1 \in \mathbb{R}$, $\Omega_2 \in \mathbb{R}$ are the upper and lower propellers' numbers of revolution per minute. Summarizing (2)–(9) yields the drone's overall attitude dynamics

$$\mathbf{I}\ddot{\mathbf{a}} = \mathbf{T}_t^b + \mathbf{T}_r^b + (\mathbf{R}\dot{\mathbf{a}})^\times \mathbf{I}\mathbf{R}\dot{\mathbf{a}} \quad (10)$$

where for an arbitrary vector $\mathbf{x} = [x_1, x_2, x_3]^\top \in \mathbb{R}^3$, $\mathbf{x}^\times := \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}$. Note that according to the symmetric structure of the drone under $\delta_{r_\phi} = \delta_{r_\theta} = 0$, the inertia matrix can be approximately calculated as $\mathbf{I} = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ where $I_{xx} \in \mathbb{R}^+$, $I_{yy} \in \mathbb{R}^+$, $I_{zz} \in \mathbb{R}^+$ are the inertia along the X, Y, Z -axes of the drone's $\sum_b$, respectively.

B. Position Kinematics and Dynamics

Denote the position vector of the drone in $\sum_i$ as $\mathbf{p} := [x, y, z]^\top \in \mathbb{R}^3$. According to mechanics of rigid bodies, the linear velocity $\dot{\mathbf{p}}$ of the drone in $\sum_i$ can be calculated by using the equation

$$\dot{\mathbf{p}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \bar{\mathbf{M}} \mathbf{v}_b = \bar{\mathbf{M}} \begin{bmatrix} v_{bx} \\ v_{by} \\ v_{bz} \end{bmatrix} \quad (11)$$

where $\mathbf{v}_b \in \mathbb{R}^3$ is the linear velocity described in $\sum_b$, and $\bar{\mathbf{M}}$ is a three-by-three transformation matrix that can be written as

$$\bar{\mathbf{M}} = \begin{bmatrix} C_\theta C_\psi & S_\phi S_\theta C_\psi - C_\phi S_\psi & C_\phi S_\theta C_\psi + S_\phi S_\psi \\ C_\theta S_\psi & S_\phi S_\theta S_\psi + C_\phi C_\psi & C_\phi S_\theta S_\psi - S_\phi C_\psi \\ -S_\theta & S_\phi C_\theta & C_\phi C_\theta \end{bmatrix}. \quad (12)$$

Then, it is straightforward that $\mathbf{v}_b$ can be calculated by

$$\mathbf{v}_b = \bar{\mathbf{M}}^{-1} \dot{\mathbf{p}} = \mathbf{M} \dot{\mathbf{p}} \quad (13)$$

where $\mathbf{M} = \bar{\mathbf{M}}^{-1}$. According to the Newton's second law, one can derive the position dynamics of the drone as

$$m \left(\frac{d\mathbf{v}}{dt} \right)_i = m \left(\frac{d\mathbf{v}}{dt} \right)_b + m (\boldsymbol{\omega}_b \times \mathbf{v}_b) \quad (14)$$

where the total force acting on the drone in the inertial frame is the sum of the forces acting on the drone in its body frame and the centripetal forces from the rotation of the body frame, $\mathbf{v} \in \mathbb{R}^3$ represents the velocity of the drone, and $(\frac{d\mathbf{v}}{dt})_i$ and $(\frac{d\mathbf{v}}{dt})_b$ are the accelerations of the drone described in $\sum_i$ and $\sum_b$, respectively. Note that the component $m(\frac{d\mathbf{v}}{dt})_b$ in (14) represents the applied force described in $\sum_b$ of the drone, which consists of two components, i.e.,

$$m \left(\frac{d\mathbf{v}}{dt} \right)_b = \mathbf{F}_t^b + \mathbf{F}_g^b \quad (15)$$

where $\mathbf{F}_t^b \in \mathbb{R}^3$ is the thrust from the propellers, and $\mathbf{F}_g^b \in \mathbb{R}^3$ is the gravity component, of which both are described in $\sum_b$. Now, we aim to obtain the specific forms for these components. First, the thrust from the propellers can be described with respect to $\sum_p$ as $\mathbf{F}_{t_p} = \begin{bmatrix} 0 \\ 0 \\ \sigma(\alpha \Omega_1^2 + \beta \Omega_2^2) \end{bmatrix}$, where $\alpha \in \mathbb{R}^+$ and $\beta \in \mathbb{R}^+$ are the lumped thrust coefficients of the upper and lower propellers, respectively. Then, we need to transfer $\mathbf{F}_{t_p}$ from $\sum_p$ to $\sum_b$ to obtain $\mathbf{F}_t^b$. According to the geometric relationship in Fig. 2, one has

$$\mathbf{F}_t^b = \sigma(\alpha \Omega_1^2 + \beta \Omega_2^2) \begin{bmatrix} S_{\delta_{r_\theta}} C_{\delta_{r_\phi}} \\ -S_{\delta_{r_\phi}} \\ C_{\delta_{r_\phi}} C_{\delta_{r_\theta}} \end{bmatrix}. \quad (16)$$

According to Fig. 2, the gravity component of the drone described in $\sum_b$ is

$$\mathbf{F}_g^b = \mathbf{M} \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} = -mg \begin{bmatrix} -\sin \theta \\ \cos \theta \sin \phi \\ \cos \theta \cos \phi \end{bmatrix}. \quad (17)$$

Summarizing (11)–(17) yields the drone's overall position dynamics

$$\begin{aligned} m\ddot{\mathbf{p}} &= \mathbf{F}_t^b + \mathbf{F}_g^b + m\boldsymbol{\omega}_b^\times \mathbf{M}\dot{\mathbf{p}} \\ &= \mathbf{F}_t^b + \mathbf{F}_g^b + m(\mathbf{R}\dot{\mathbf{a}})^\times \mathbf{M}\dot{\mathbf{p}} \end{aligned} \quad (18)$$

where $\mathbf{M}$ defined in (13) is always nonsingular, which makes (18) well-defined.

C. Analysis of the Overall Six DoF Dynamics

From the last section, the overall attitude and position dynamics can be written in a compact form as

$$\begin{cases} I\ddot{\mathbf{a}} - \mathbf{T}_t^b - \mathbf{T}_r^b - (\mathbf{R}\dot{\mathbf{a}})^\times I\mathbf{R}\dot{\mathbf{a}} = 0 \\ m\ddot{\mathbf{p}} - \mathbf{F}_t^b - \mathbf{F}_g^b - m(\mathbf{R}\dot{\mathbf{a}})^\times M\dot{\mathbf{p}} = 0. \end{cases} \quad (19)$$

Now, we analyze the properties of the overall dynamics (19), which will be the foundation for the control design.

1) *Underactuated Dynamics*: In the overall nonlinear dynamics (19), the states $[\mathbf{a}^\top, \mathbf{p}^\top]^\top \in \mathbb{R}^6$ are six dimensions. However, the control input $[\Omega_1^2, \Omega_2^2, \delta_{r\theta}, \delta_{r\phi}] \in \mathbb{R}^4$ are only four dimensions, which implies an underactuated dynamics. Therefore, we are only able to fully control four states. As similar to the quadrotor case, the three DoF positions and yaw angle are usually more of importance in many applications. Therefore, we can select them as the four controllable states, and firstly leave roll and pitch angles as uncontrollable states.

2) *Coupling Effect*: The attitude dynamics in (19) are independent to the position dynamics, while the position dynamics are coupled with the attitude dynamics due to the existence of the component $m(\mathbf{R}\dot{\mathbf{a}})^\times M\dot{\mathbf{p}}$. Due to the absence of an attitude damping part in (10), the attitude dynamics are naturally unstable in the sense that given a nonzero initial states, the attitude may diverge or oscillate without external control torques. This can be verified by linearizing the attitude dynamics around the equilibrium $\dot{\mathbf{a}} = 0$. Thus, additional damping components are needed to stabilize the attitude dynamics. In addition, due to the existence of the gravity component $\mathbf{F}_g^b$ in the position dynamics, the control force $\mathbf{F}_t^b$ should not be zero even at a stable state.

3) *Model Accuracy*: In this model, the drone has been designed such that its CG lies at the center of its fuselage section. This design specification can be achieved by moving the heavier components in the drone into the fuselage section. In addition, according to the later simulation and experiments, the drone keeps almost vertical during hovering and slowly maneuvering missions, for which the current modeling based on static CG can be used in many scenarios. While for fast maneuvering mission, the modeling based on multibody dynamics might be more appropriate, which will be considered as our future work.

D. Comparison of Overall Dynamics With Quadrotors

We provide a complete comparison between the coaxial drone and quadrotors from the perspective of motion dynamics. By letting the common parameters the same, such as the air density ρ and the drones' mass m , Table I summarizes the main differences between these two types of drones. According to Table I, the produced thrust along roll or pitch axis is obtained by projecting the overall thrust to the roll or pitch axis. Therefore, under the common parameters and allowed tilting angles, the coaxial drone's thrust along the roll or pitch axis is larger than the corresponding one produced by quadrotors. In addition, to accelerate along the pitch axis, quadrotors need to adjust their attitudes to incline first, then use the projected thrust to produce the desired motion in the pitch axis. Different from quadrotors, the coaxial drone can produce the thrust along the pitch axis

TABLE I
COMPARISON OF THE COAXIAL DRONE AND QUADROTORS

Property	Drone	Coaxial drone	Quadrotors
Translational dynamics		$m\ddot{\mathbf{p}} - \mathbf{F}_t^b - \mathbf{F}_g^b$ $-m(\mathbf{R}\dot{\mathbf{a}})^\times M\dot{\mathbf{p}} = 0$	$m\ddot{\mathbf{p}} + m\mathbf{g}\mathbf{e}_z$ $= \mathbf{R}_b^i \mathbf{F}_t^b$
Rotational dynamics		$I\ddot{\mathbf{a}} - \mathbf{T}_t^b - \mathbf{T}_r^b$ $-(\mathbf{R}\dot{\mathbf{a}})^\times I\mathbf{R}\dot{\mathbf{a}} = 0$	$\mathbf{J}\ddot{\mathbf{a}} + \mathbf{C}(\mathbf{a}, \dot{\mathbf{a}})\dot{\mathbf{a}}$ $= \mathbf{T}_t^b$
Control inputs		$\Omega_1, \Omega_2, \delta_{r\phi}, \delta_{r\theta}$	$\omega_1, \omega_2, \omega_3, \omega_4$
Produced thrust along yaw axis		$2\rho C_T \pi \omega^2 d_r^4$ $\times C_{\delta_{r\theta}} C_{\delta_{r\phi}}$	$\rho C_T \pi \omega^2 d_r^4 / 4$ $\times C_\phi C_\theta$
Produced thrust along roll axis		$2\rho C_T \pi \omega^2 d_r^4$ $\times C_{\delta_{r\theta}} S_{\delta_{r\phi}}$	$\rho C_T \pi \omega^2 d_r^4 / 4$ $\times (C_\phi S_\theta C_\psi + S_\phi S_\psi)$
Produced thrust along pitch axis		$2\rho C_T \pi \omega^2 d_r^4$ $\times S_{\delta_{r\phi}}$	$\rho C_T \pi \omega^2 d_r^4 / 4$ $\times (S_\phi S_\theta C_\psi - S_\phi C_\psi)$
Produced force in body frame ($\mathbf{F}_t^b$)		$\sigma(\alpha\Omega_1^2 + \beta\Omega_2^2)$ $\times \begin{bmatrix} S_{\delta_{r\theta}} C_{\delta_{r\phi}} \\ -S_{\delta_{r\phi}} \\ C_{\delta_{r\theta}} C_{\delta_{r\phi}} \end{bmatrix}$	$\begin{bmatrix} 0 \\ 0 \\ c \sum_{i=1}^4 \omega_i^2 \end{bmatrix}$
Produced torque in body frame ($\mathbf{T}_t^b$)		$\sigma(\gamma_1\Omega_1^2 - \gamma_2\Omega_2^2)$ $\times \begin{bmatrix} S_{\delta_{r\theta}} C_{\delta_{r\phi}} \\ -S_{\delta_{r\phi}} \\ C_{\delta_{r\theta}} C_{\delta_{r\phi}} \end{bmatrix}$	$\begin{bmatrix} c(\omega_1^2 - \omega_2^2) \\ c(\omega_2^2 - \omega_4^2) \\ \bar{c}(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \end{bmatrix}$

directly by tilting the lower servomotor, which is also shown in Table I's second line from the bottom.

IV. CONTROL OF THE COAXIAL DRONE

As stated in Section III-C, we aim to control p and ψ using $\Omega_1^2, \Omega_2^2, \delta_{r\theta}, \delta_{r\phi}$. Due to the high nonlinear property of the dynamics (19), we first design a desired control force and yaw torque to stabilize $\mathbf{p}$ and ψ , then allocate the desired control force and yaw torque to the four real control inputs.

A. Desired Control Input

For the attitude dynamics, consider a desired yaw angle $\psi^d(t) \in \mathbb{R}$, which is continuous and twice differentiable. Then, to make $\psi(t)$ track $\psi^d(t)$, the desired control yaw torque can be designed as

$$\begin{aligned} T_{\psi d} = & - \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}^\top (\mathbf{R}\dot{\mathbf{a}})^\times I\mathbf{R}\dot{\mathbf{a}} \\ & + I_{zz} [\ddot{\psi}^d - k_d(\dot{\psi} - \dot{\psi}^d) - k_p(\psi - \psi^d)] \end{aligned} \quad (20)$$

where $k_d > 0, k_p > 0$ are velocity feedback and state feedback gains, respectively.

For the position dynamics, consider a desired flight trajectory $\mathbf{p}^d(t) \in \mathbb{R}^3$, which is continuous and twice differentiable. Then, based on the fully actuated system approaches [21], the desired control force can be designed as

$$\begin{aligned} \mathbf{F}_d = & -\mathbf{F}_g^b - m(\mathbf{R}\dot{\mathbf{a}})^\times M\dot{\mathbf{p}} \\ & + m [\ddot{\mathbf{p}}^d - k_d(\dot{\mathbf{p}} - \dot{\mathbf{p}}^d) - k_p(\mathbf{p} - \mathbf{p}^d)]. \end{aligned} \quad (21)$$

Then, we have the following results.

Theorem 1: For the coaxial drone whose dynamics are modeled as (19), if the control inputs $\Omega_1^2, \Omega_2^2, \delta_{r_\theta}, \delta_{r_\phi}$ are designed such that

$$[0, 0, 1](\mathbf{T}_t^b + \mathbf{T}_r^b) = T_{\psi d}, \quad \mathbf{F}_t^b = \mathbf{F}_d \quad (22)$$

then the position error $\tilde{\mathbf{p}}(t) = \mathbf{p}(t) - \mathbf{p}^d(t)$ and yaw angle error $\tilde{\psi}(t) = \psi(t) - \psi^d(t)$ converge to zero.

Proof: Substituting the controllers (20)–(22) into the overall dynamics (19) yields

$$\begin{cases} m\ddot{\tilde{\mathbf{p}}}(t) + mk_d\dot{\tilde{\mathbf{p}}}(t) + mk_p\tilde{\mathbf{p}}(t) = 0 \\ I_{zz}\ddot{\tilde{\psi}}(t) + I_{zz}k_d\dot{\tilde{\psi}}(t) + I_{zz}k_p\tilde{\psi}(t) = 0. \end{cases} \quad (23)$$

Since $m > 0$ and $I_{zz} > 0$, the characteristic polynomial of the linear dynamics (23) can be written as

$$s^2 + k_d s + k_p = 0 \quad (24)$$

where $s \in \mathbb{C}$. Since $k_d > 0$ and $k_p > 0$, $\tilde{\mathbf{p}}(t)$ and $\tilde{\psi}(t)$ converge to zero. $\square$

B. Control Allocation

Note that Theorem 1 relies on the assumption that (22) should hold. In this section, we investigate how to design control inputs $\Omega_1^2, \Omega_2^2, \delta_{r_\theta}, \delta_{r_\phi}$ such that (22) holds, which is inherently a control allocation problem.

First, substituting the definitions of $\mathbf{T}_t^b$ and $\mathbf{T}_r^b$ in (5)–(9) into (22) yields

$$\begin{aligned} T_{\psi d} = & \sigma \left[C_{\delta_{r_\theta}} C_{\delta_{r_\phi}} (\gamma_1 \Omega_1^2 - \gamma_2 \Omega_2^2) \right. \\ & \left. + d_2 S_{\delta_{r_\theta}} S_{\delta_{r_\phi}} (\alpha \Omega_1^2 + \beta \Omega_2^2) \right]. \end{aligned} \quad (25)$$

Then, substituting the definition of $\mathbf{F}_t^b$ in (16) into (22) yields

$$F_d(1) = \sigma S_{\delta_{r_\theta}} C_{\delta_{r_\phi}} (\alpha \Omega_1^2 + \beta \Omega_2^2) \quad (26)$$

$$F_d(2) = -\sigma S_{\delta_{r_\phi}} (\alpha \Omega_1^2 + \beta \Omega_2^2) \quad (27)$$

$$F_d(3) = \sigma C_{\delta_{r_\theta}} C_{\delta_{r_\phi}} (\alpha \Omega_1^2 + \beta \Omega_2^2). \quad (28)$$

Note that $T_{\psi d}, F_d(1), F_d(2), F_d(3)$ are known from the right-hand side of (20) and (21). Therefore, we need to find a unique mapping from $\Omega_1^2, \Omega_2^2, \delta_{r_\theta}, \delta_{r_\phi}$ to $T_{\psi d}, F_d(1), F_d(2), F_d(3)$ such that (25)–(28) holds.

Different from linearization-based control approach [14], we propose a precisely nonlinear control allocation approach here. Dividing (26) by (28) yields

$$\delta_{r_\theta} = \arctan \frac{F_d(1)}{F_d(3)}. \quad (29)$$

Dividing (27) by (28) yields

$$\delta_{r_\phi} = -\arctan \frac{F_d(2)C_{\delta_{r_\theta}}}{F_d(3)} \quad (30)$$

where $C_{\delta_{r_\theta}}$ can be known from (29). After the knowledge of δ_{r_θ} and δ_{r_ϕ} , (28) can be rewritten as

$$\alpha \Omega_1^2 + \beta \Omega_2^2 = \frac{F_d(3)}{\sigma C_{\delta_{r_\theta}} C_{\delta_{r_\phi}}}. \quad (31)$$

Substituting (31) into (25) yields

$$\gamma_1 \Omega_1^2 - \gamma_2 \Omega_2^2 = \frac{T_{\psi d} C_{\delta_{r_\theta}} C_{\delta_{r_\phi}} - F_d(3) d_2 S_{\delta_{r_\theta}} S_{\delta_{r_\phi}}}{\sigma C_{\delta_{r_\theta}}^2 C_{\delta_{r_\phi}}^2}. \quad (32)$$

Combining (31) and (32) yields

$$\begin{aligned} \Omega_1^2 &= \frac{\gamma_2 F_d(3) C_{\delta_{r_\theta}} C_{\delta_{r_\phi}} + \beta T_{\psi d} C_{\delta_{r_\theta}} C_{\delta_{r_\phi}} - \beta F_d(3) d_2 S_{\delta_{r_\theta}} S_{\delta_{r_\phi}}}{\sigma C_{\delta_{r_\theta}}^2 C_{\delta_{r_\phi}}^2 (\alpha \gamma_2 + \beta \gamma_1)} \\ \Omega_2^2 &= \frac{\gamma_1 F_d(3) C_{\delta_{r_\theta}} C_{\delta_{r_\phi}} - \alpha T_{\psi d} C_{\delta_{r_\theta}} C_{\delta_{r_\phi}} + \alpha F_d(3) d_2 S_{\delta_{r_\theta}} S_{\delta_{r_\phi}}}{\sigma C_{\delta_{r_\theta}}^2 C_{\delta_{r_\phi}}^2 (\alpha \gamma_2 + \beta \gamma_1)}. \end{aligned} \quad (33)$$

$$\Omega_1^2 = \frac{\gamma_1 F_d(3) C_{\delta_{r_\theta}} C_{\delta_{r_\phi}} - \alpha T_{\psi d} C_{\delta_{r_\theta}} C_{\delta_{r_\phi}} + \alpha F_d(3) d_2 S_{\delta_{r_\theta}} S_{\delta_{r_\phi}}}{\sigma C_{\delta_{r_\theta}}^2 C_{\delta_{r_\phi}}^2 (\alpha \gamma_2 + \beta \gamma_1)}. \quad (34)$$

Now, we summarize the control allocation steps as the following results.

Proposition 1: Given the desired yaw torque $T_{\psi d}$ and force $\mathbf{F}_d$ and the relationship (25)–(28), the four control inputs can be uniquely allocated using (29), (30), (33), and (34). Moreover, if we command the coaxial drone according to (29), (30), (33), and (34), then the tracking errors $\tilde{\mathbf{p}}(t) = \mathbf{p}(t) - \mathbf{p}^d(t)$ and $\tilde{\psi}(t) = \psi(t) - \psi^d(t)$ converge to zero.

Since all the calculation steps from (29)–(34) are unique, it is straightforward to have the proof of Proposition 1.

Remark 3: In (29), (30), (33), and (34), due to the existence of denominator parts in the calculations of the control inputs $\Omega_1^2, \Omega_2^2, \delta_{r_\theta}, \delta_{r_\phi}$, it is very important that the calculation of (29), (30), (33), and (34) should not have singularities. Since the drone's gravity force needs to be canceled by $F_d(3)$, the magnitude of $F_d(3)$ will be sufficiently away from zero, which makes the calculation (29)–(30) singularity-free. Also, the desired tilt angles will be zero when the drone is hovering and be bounded when the drone is maneuvering, which makes the calculation (33) and (34) also singularity-free.

C. Damping Uncontrolled Roll and Pitch Dynamics

After the position and yaw angle of the drone are controlled to track their desired trajectory and yaw angle, the dynamics of the roll and pitch angles will be fully uncontrolled and evolve according to (19). According to (20) and (21), the divergence of the roll and pitch angular velocities will also make the whole system divergent since both $T_{\psi d}$ and $\mathbf{F}_d$ will become infinite, thus making $\Omega_1^2, \Omega_2^2, \delta_{r_\theta}, \delta_{r_\phi}$ also infinite. Therefore, to avoid this scenario, suppressing the roll and pitch angular velocities is *crucial*.

We note that the vibration of the roll and pitch angles of the drone originates from the actions applied to the two tilt servomotors. Therefore, a natural solution is to add roll and pitch damping components into the corresponding servomotors. More specifically, instead of commanding the two servomotors according to (29) and (30), we command them by

$$\delta_{r_\theta} = \arctan \frac{F_d(1)}{F_d(3)} + k_\theta \dot{\theta} \quad (35)$$

$$\delta_{r\phi} = -\arctan \frac{F_d(2)C_{\delta_{r\theta}}}{F_d(3)} + k_\phi \dot{\phi} \quad (36)$$

where k_θ and k_ϕ are gain scalars, which can be tuned according to the performance of experimental tests. We will show in the next simulation and experimental sections that the damping components in (35) and (36) are very efficient to stabilize the uncontrolled roll and pitch angular velocities.

Remark 4: Different from the controller designed in [12] where the linearization technique is used to analyze the closed-loop dynamics, our proposed controller uses a model-based control allocation approach, where linearization is not needed. Also, different from the nonlinear control approaches for coaxial drones [12], [22], a novel nonlinear control allocation-based approach is proposed for the designed coaxial drone.

Remark 5: There are four control gains $k_p, k_d, k_\theta, k_\phi$ in the proposed controllers (20), (21), (35), and (36). Indeed, different selection of control gains will give different system performance, for which we will tune the gain parameters first in our experiments. Since the position and attitude dynamics (19) are second-order, the system performance may not be the best if the gains are too large or small. In addition, the two servomotors' updating frequency also determines the system performance, where the difference is that the performance is usually better given larger updating frequency.

Remark 6: In the proposed controllers (20), (35), and (36), the angular velocities in roll, pitch, and yaw axes are needed, which are usually noisy if they are obtained from a pure inertial measurement unit (IMU). To improve IMUs' measuring accuracy and reduce noise's magnitude, many auxiliary measurements and techniques have been developed, such as the sensor fusion based on magnetic measurements and advanced digital filtering techniques. Therefore, we can get the angular velocities with relatively small magnitude of noise from these IMUs. In addition, from the dynamics of the drone, (19) indicates that the system performance depends on the control laws' PD gain parameters, the damping gains, and the updating frequency of the servomotors, etc. Therefore, by experimental tests, we need to choose better control gains to mitigate the negative effect of measurement noises.

V. NUMERICAL SIMULATIONS

In this section, we conduct two types of simulations, namely mathematical model simulation and physical model simulation, to verify the established model and validate the effectiveness of the proposed control strategy. The related parameters of an actual manufactured coaxial drone are listed in the following, which are also used in all the simulation examples in this section: mass $m = 652.3$ g, $d_1 = 67.1$ mm, $d_2 = 67.2$ mm, $d_3 = 46.6$ mm, and inertial matrix $\mathbf{I} = \begin{bmatrix} 5.83 \times 10^6 & 0 & 0 \\ 0 & 5.64 \times 10^6 & 0 \\ 0 & 0 & 3.79 \times 10^5 \end{bmatrix} \text{ g} \cdot \text{mm}^2$ (obtained from the Solidworks software), $\gamma_1 = 0.000182$, $\gamma_2 = 0.000207$, $\alpha = 0.01227$, $\beta = 0.01261$, and $\sigma = 0.673387$ (obtained from experimental tests which will be introduced in next section). The damping gains are selected as $k_\theta = 0.03$, $k_\phi =$

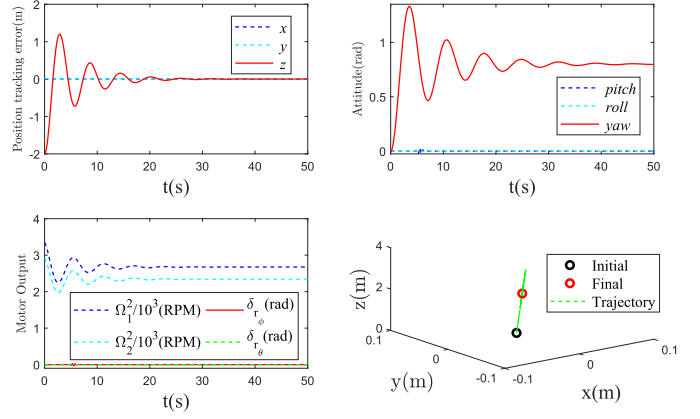


Fig. 4. Mathematical model simulation for the hovering case with damping (35) and (36) (MATLAB).

0.03, which are used in all the following simulations and the flight experiments. The initial states are $\mathbf{p}(0) = [0, 0.01, 0]^T$ and $\mathbf{a}(0) = [0, 0.01, 0]^T$. The video showing the simulation results and the designed coaxial drone's real flight tests has been uploaded in the supplementary materials, available online, due to space limitations.³

A. Mathematical Model Simulation in MATLAB

This simulation is conducted upon the established mathematical model (19). First, we consider a hovering mission for the drone by assigning the desired position as $\mathbf{p}^d = [0, 0, 2]^T$ m and desired yaw angle as $\psi^d = 0.8$ rad. Under the gain parameters $k_p = 0.8$, $k_d = 0.23$ and the damping terms (35) and (36), the simulation results are shown in Fig. 4. From the position tracking error and the evolution of the drone's attitude, one can see that the desired position and desired yaw angle are achieved within about 25 s. The four motors' output shows that each propeller will produce about $mg/2[N]$ thrust when the position of the drone reaches its destination. The trajectory of the drone is also shown in the last subfigure of Fig. 4. To show the effect of the proposed damping terms in (35) and (36), we also conduct a mathematical model simulation for the hovering case with the same gain parameters but without the damping terms (35) and (36). According to the simulation results in Fig. 5, although the position tracking error, pitch angle and roll angle become smaller, the yaw angle is unstable and the output of the servomotors is unstable. This validates that the proposed damping terms (35) and (36) are very important in the stabilization of the drone's attitude.

Then, we consider a maneuvering mission for the drone by assigning the desired position as $\mathbf{p}^d(t) = [\sin t, \cos t, 1]^T$ m and desired yaw angle as $\psi^d = 0.8$ rad. By using the gain values $k_p = 0.8$, $k_d = 2$, the simulation results are shown in Fig. 6. For the maneuvering case, one can also see that the desired position and yaw angle are achieved within about 20 s, where small tracking errors exist in x and y directions. Note that when the drone

³[Online]. Available: https://www.bilibili.com/video/BV1x5411y7g9/?spm_id_from=333.999.0.0

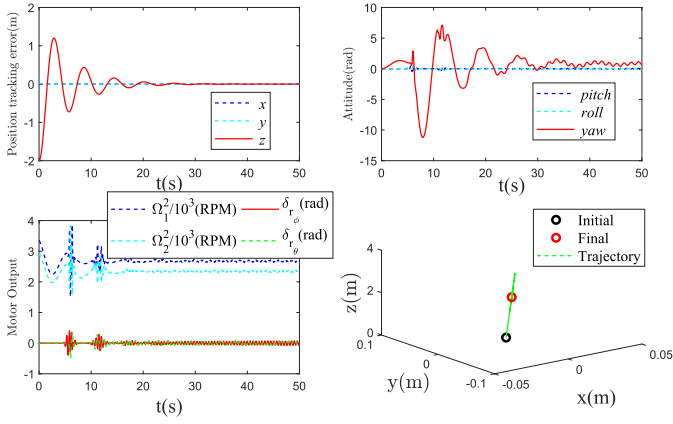


Fig. 5. Mathematical model simulation for the hovering case without damping (35)–(36) (MATLAB).

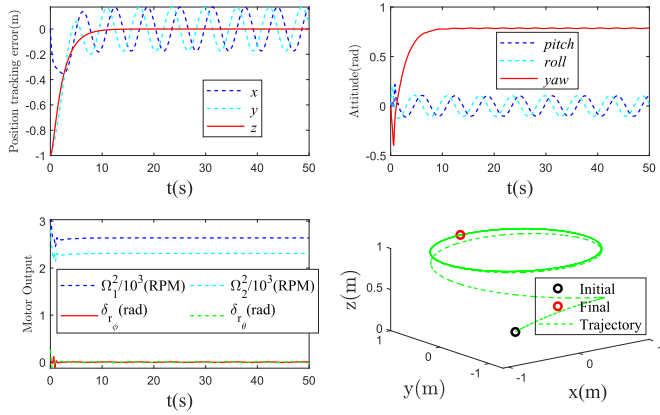


Fig. 6. Mathematical model simulation for the maneuvering case (MATLAB).

achieves the maneuvering mission, the roll and pitch angles change periodically, and the outputs of the two servomotors also change periodically. This is because the desired maneuvering trajectory changes periodically. The last subfigure of Fig. 6 shows that the desired trajectory is achieved by the drone. This validates the effectiveness of the proposed control strategy for the coaxial drone in tracking a time-varying trajectory.

B. Physical Model Simulation in Unreal Engine

This simulation is conducted on the physical drone model established in Unreal Engine software, which is exported from Solidworks. We also consider hovering and maneuvering tasks with the same desired trajectories. To compare with the mathematical simulation, we use the same PD gain values in this simulation. However, to eliminate constant convergence error, we employ PID in this simulation, where the gain for the integral term is 0.98. The simulation results for the hovering case are provided in Fig. 7, where the output of servomotors, the evolution of the flight trajectory and position tracking error are similar between Figs. 4 and 7.

The simulation results for the maneuvering case are provided in Fig. 8. Compared to Fig. 6, the flight trajectory and the output

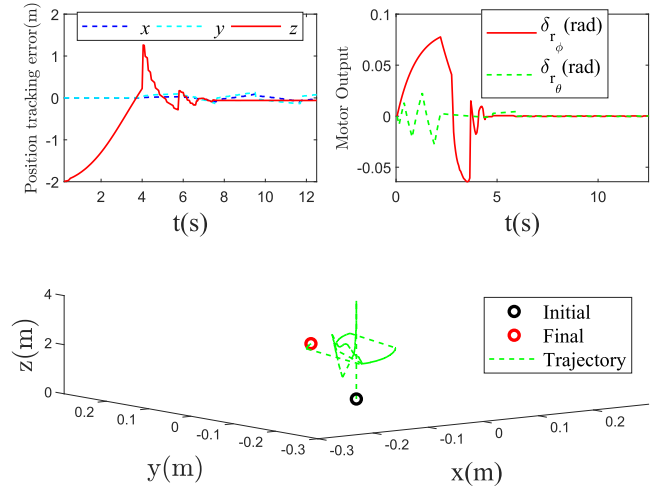


Fig. 7. Physical model simulation for the hovering case (Unreal Engine).

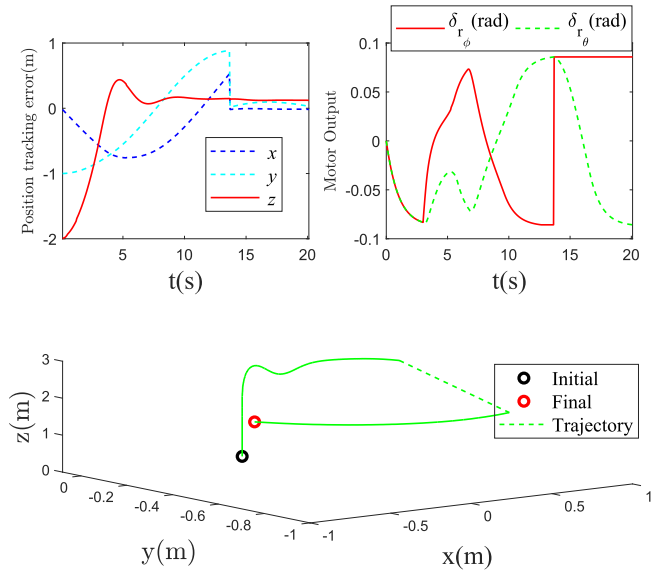


Fig. 8. Physical model simulation for the maneuvering case (Unreal Engine).

of the servomotors are similar in Fig. 8. These simulation results validate that the established mathematical model of the drone is close to the physical model of the drone in the simulation environment, in which their difference is mainly caused by the modeling error.

VI. REAL-FLIGHT EXPERIMENTS

In this section, we conduct four classes of experimental tests on our physical coaxial drone to validate its design and the proposed control strategy. The first two classes of experiments aim to validate the design by testing the coaxial propellers' thrust efficiency and comparing the thrust produced by the coaxial drone and quadrotors. The latter two classes of experiments aim to validate the proposed control strategy by testing the effect of the designed damping terms and testing the hovering flight. Note that the values of thrust efficiency and lumped thrust and

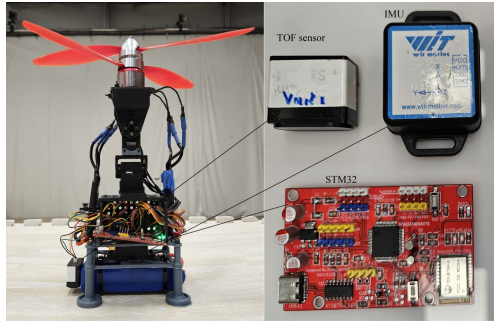


Fig. 9. Designed coaxial drone and its main components.

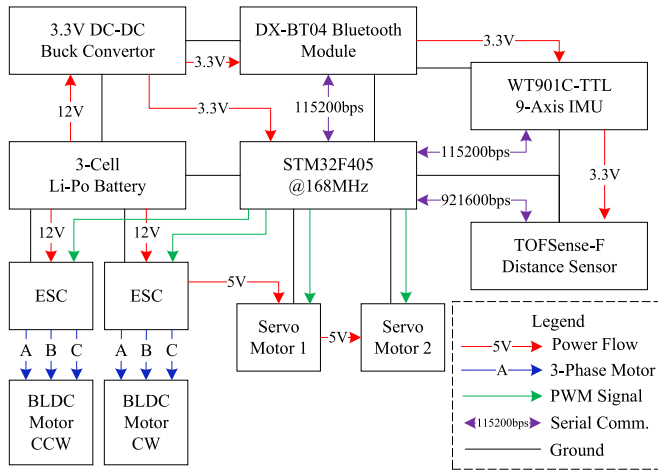


Fig. 10. Graphical representation of components' connection.

moment coefficients are obtained in Section VI-E, which are used in the aforementioned simulation examples. The gain values employed in the follow-up experiments are the same as those in the previous MATLAB simulation. In the following sections, we first introduce the system framework and the components' connection, then introduce these four experiments, respectively.

A. System Framework and Components' Connection

As shown in Fig. 9, the physical drone follows the exact design introduced in Section II, where four components are stacked in the fuselage, including battery, STM32 micro controller unit (MCU), IMU sensor, and time of flight (TOF) sensor (which can be used to measure the flight height during each updating interval).

As shown in Fig. 10, the electronic components in the drone are organized in a manner to optimize overall flow of power and signals' connection. Specifically, the coaxial drone is powered by a 3S Li-Po battery, where voltage regulators are employed to obtain suitable operating voltages for other components. The flight controller is implemented upon the MCU STM32F405RGT6 at a core frequency of 168 MHz. The codes inside the MCU are developed and debugged on VSCODE and PlatformIO IDE with C++. The MCU communicates with the host computer through the DX-BT04 Bluetooth module at a baud rate of 115 200 bps via the channel USART1, where the

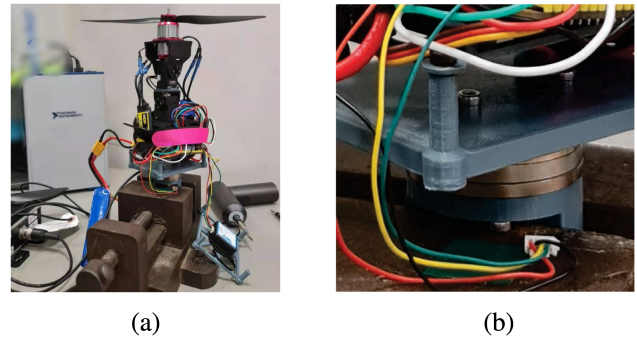


Fig. 11. (a) Testing platform. (b) Force sensor.

transmitted data consists of control commands (such as takeoff and landing) and preset parameters. IMU data are transmitted by the channel USART2 under an interrupt mode at a baud rate of 115200 bps. TOF sensor data are transmitted by the channel USART3 under an interrupt mode at a baud rate of 921600 bps. The control algorithm proposed in this article is executed under the timer interrupt with an updating frequency of 250 Hz. The magnitude of the four control outputs are represented by pulse width modulation (PWM) signals with a common frequency of 100 Hz, which are used to control the propellers and servomotors.

The MCU sets four interrupt priority levels, where the order is USART1, timer interrupt, USART3, and USART2. After the drone is powered on, it enters the initialization mode and waits for a JSON-formatted string input to adjust target height, PID parameters, damping ratio, servo sensitivity, and other parameters. After initialization is complete, the drone will fly autonomously according to the calculated control commands. During flight, the MCU will also upload the control commands and the sensing data obtained from, e.g., IMU to the host computer via the Bluetooth at a frequency of 25 Hz.

B. Experimental Tests on the Thrust Efficiency

The coaxial drone's thrust efficiency plays an important role in Sections II-E and III-A, for which we now design experiments to test its value. The testing platform is shown in Fig. 11(a). We have used a high-precision sensor, ATI DAQ Net F/T⁴, to measure the force and torque produced by the drone. The upper side of the sensor is rigidly attached to the bottom of the drone, and the bottom side of the sensor is fixed by a ground station, as shown in Fig. 11(b). Using the software from the company ATI, we could read the force and torque applied to the drone, which is produced by the rotation of the propellers. Fig. 12 shows the testing data and the fitting relation from the PWM signal (produced from electronic speed controller, ESC) to the produced thrust. The existence of energy loss is clearly shown in Fig. 12 since the sum of the individual propeller's produced thrust is higher than the thrust produced by the coaxial propellers. Using these testing data, the energy loss coefficients of the thrust is approximated as $\sigma \approx \frac{0.0167}{0.0122+0.0126} \approx 67.3\%$, in which we use the ESC's PWM value to replace the propellers' rotating speed in

⁴[Online]. Available: <https://www.ati-ia.com/products/ft/sensors.aspx>

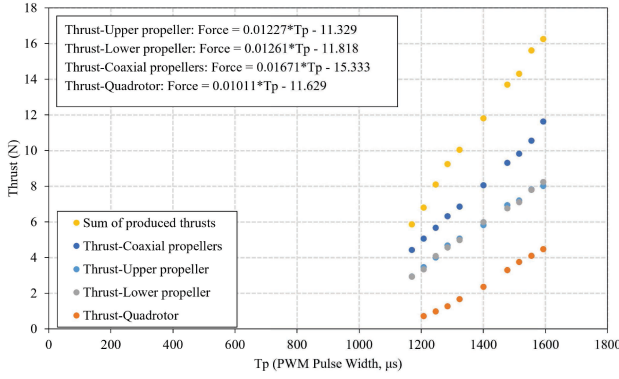


Fig. 12. Testing results on the thrust produced by the coaxial drone and the quadrotor.

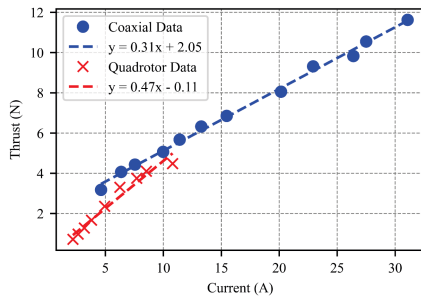


Fig. 13. Testing results on the thrust and current produced by the coaxial drone and quadrotor.

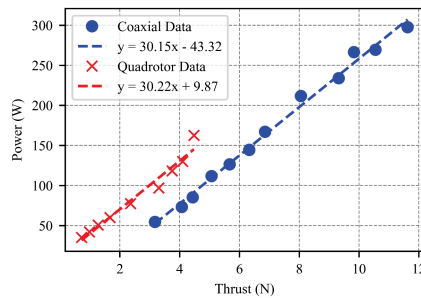


Fig. 14. Testing results on the thrust and power produced by the coaxial drone and quadrotor.

this calculation since the propellers' rotating speed is unavailable and the rotation of the propellers are controlled by the ESC. To further show the thrust efficiency, we also show the testing results on the drone's produced current, thrust, and power in Figs. 13 and 14. The current and power are measured by a specialized instrument MDA 8000HD Motor Drive Analyzers.

C. Thrust Comparison Between the Coaxial Drone and a Quadrotor

To validate the comparison in Section II-E, we conduct real experiments on the coaxial drone and a quadrotor whose propeller size is half of that of the coaxial drone, as shown in Fig. 15. The experiments are also conducted on the testing platform introduced in Fig. 11(a). To have a fair comparison,

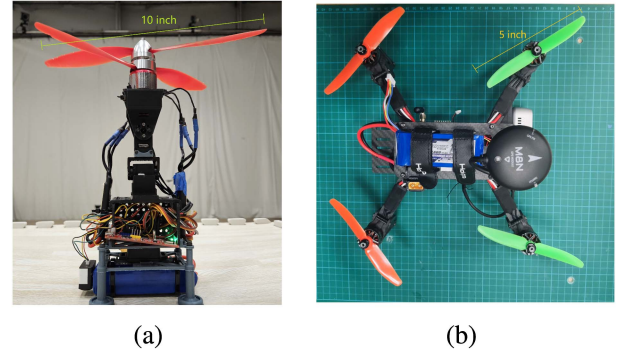


Fig. 15. (a) Coaxial drone. (b) Quadrotor.

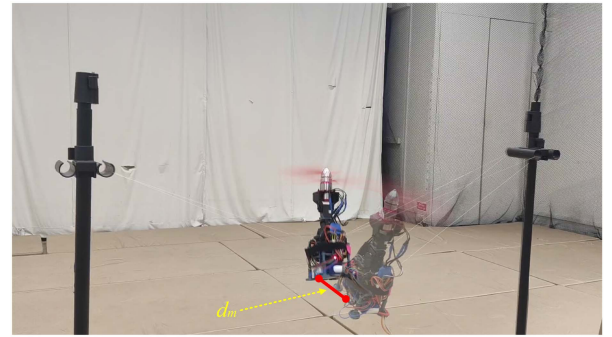


Fig. 16. Testing the effect of the damping gains and motors' updating frequency.

the other testing conditions, parameters and devices are kept the same as much as possible, such as the ESC. The testing results are shown in Fig. 12, which indicates that the coaxial drone's produced thrust is always larger than that of the quadrotor. Using these data given in Fig. 12, the range of $\frac{\text{Thrust-Coaxial propellers}}{\text{Thrust-Quadrotor}}$ can be calculated, which is (2.75, 9.38). Indeed, the corresponding theoretical ratio $\frac{1.3}{1/4} = 5.2$ given in Section II-E is in this experimental range. Furthermore, the comparison of current and power for the coaxial drone and quadrotor is provided in Figs. 13 and 14, from which one has that with the same produced thrust, the quadrotor needs higher current and power than that of the coaxial drone.

D. Experimental Tests on the Damping Gains and Motors' Updating Frequency

As introduced in Remark 5, the four control gains and motors' updating frequency determine the coaxial drone's system performance. Thus, our intent in this section is to obtain the best control gains and updating frequency for flights by testing the system response under different parameters. As shown in Fig. 16, when the drone's initial height is exactly or very close to the desired height (realized by suspending the drone with two lines connected with two ground poles), the drone's motion along yaw axis is minor. Then, the oscillation of the drone along the roll axis describes the system performance of the drone's dynamics along the roll axis (the motion along pitch axis is restricted by the lines). Therefore, as shown in Fig. 16, we can use the maximum distance d_m of the

TABLE II
EFFECT OF DAMPING GAINS AND UPDATING FREQUENCY

Parameters	Properties	Tested values	Response d_m
Damping gains $k_\theta = k_\phi$	0	10 cm	
	0.03	5 cm	
	0.15	Unstable	
Motors' updating frequency	25 Hz	30 cm	
	50 Hz	20 cm	
	250 Hz	5 cm	

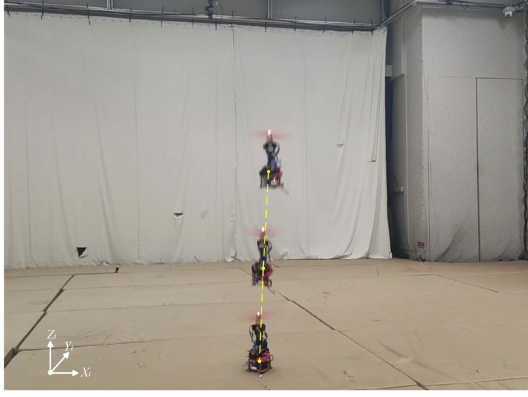


Fig. 17. Vertical ascending flight.

drone's motion in the horizontal plane to approximately characterize the system performance of the drone's roll dynamics. Then, one has $d_m \approx |\max x[k] - \min x[k]|, \forall k, k = 1, 2, \dots$, which can be obtained from our on-board sensor measurements, specifically the motion distance and the drone's attitude. The testing results are summarized in Table II. From the table, the best parameters are $k_\theta = 0.03, k_\phi = 0.03, f_{\text{servo}} = 250\text{Hz}$ since the oscillation is minimum among these cases.

E. Experimental Tests on Vertical Ascending Flight

In this experiment, we conduct vertical ascending flight where the drone is commanded to reach a target height. Indeed, to achieve such mission, accurate height measurements are needed, which is based on the fusion of the TOF and IMU measurements since TOF sensor measures motion distance and IMU measures the drone's attitude. The drone's flight results are shown in Figs. 17 and 18. To validate our proposed control strategy, we developed our own flight control platform for the coaxial drone. Due to the absence of global positioning system, the maneuvering flight task is challenging, for which we tested the maneuvering flight under a manual control mode. The flight results under these cases can be found in the video.

Meanwhile, based on the analysis of the experimental data in the above-mentioned ascending flight, it is observed that the drone became less stable when it operates in the air with longer time. Specifically, after the drone reaches the desired position, it gradually become less stable. This is mainly caused by the measurement inaccuracy (or error accumulation) of the drone's state from IMU, which is directly included in the drone's control algorithm. Therefore, to avoid this issue, our platform is also tested based on the advanced sensor fusion and filtering

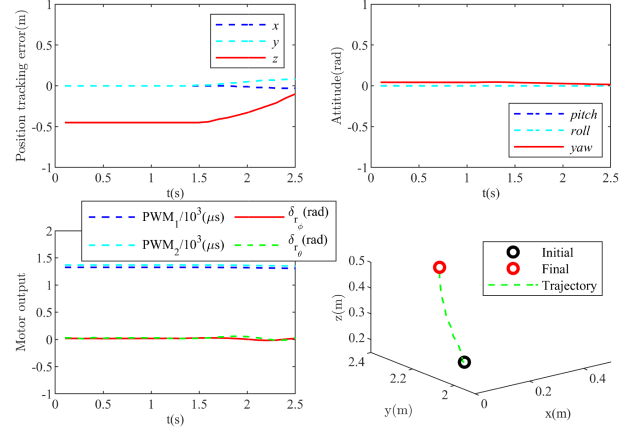


Fig. 18. Experimental data in the vertical ascending flight.

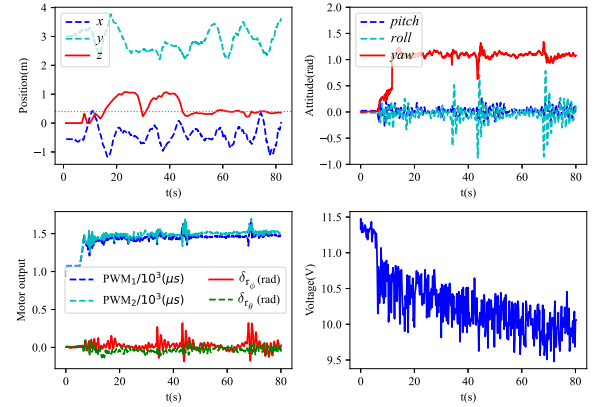


Fig. 19. Experimental data in the hover flight.

techniques embedded in Pixhawk 4 mini, in which the existing control allocation framework is employed. The flight results are shown in Fig. 19 and the video. According to Fig. 19, the attitude and position of the drone are almost maintained, and the motors' output is relatively stable during the hover. This further demonstrates the effective design of the drone and highlights the contributions of the experiments.

Comparing the experimental data with the simulation data, one has that there exist common aspects and also differences. The common aspects include the effect of the damping components and the trend of the control outputs, while the differences include the employed gain values and the system evolution. The reason of their difference is caused by the modeling inaccuracy, for which the best gain for the simulation example might not the best for the experiment. Although there exist differences, the established mathematical model helps us understand the drone and propose a controller, and the simulations help us verify them rapidly.

VII. CONCLUSION AND FUTURE WORK

In this article, we have designed an innovative coaxial drone such that the advantages of maximum thrust per platform area and flight maneuverability can be simultaneously actualized using thrust-vectoring coaxial propellers. The contra-rotating coaxial propellers can be actuated using a left-right tilt

servomotor and a back–forward tilt servomotor connected vertically in series. The drone’s position and attitude dynamics with six DoF have been modeled, which is underactuated since the number of the control inputs is only four. An efficient controller with a nonlinear control allocation has been proposed. For the uncontrolled roll and pitch dynamics, a damping part has been delicately added such that the roll and pitch angular velocities can also be stabilized. Both numerical simulations and real experiments have been conducted, which have validated the effectiveness of the drone design, the established model and the proposed control strategy.

An interesting direction for future work is the usage of the coaxial drone for aerial manipulation. Specifically, if an one-link arm is installed at the bottom side of the drone and the other side of the arm is with adhesive/magnetic mechanisms, then the coaxial drone can be used to pick up a target and move it to a destination. In such case, the CG of the combined arm and drone is shifted downward, as compared with the original drone. If the connection between the arm and the target is rigid, then the arm and the target can be regarded as a part of the drone’s fuselage part, in which our proposed control design still applies. For the case of a two-link arm, if the two links are not vertical, then the assumption regarding the CG of the whole structure may be violated. An alternative for such case is to use the Lagrangian approach to model the dynamics of the drone. Specifically, we need to calculate the kinetic and potential energy of the drone consisting of multibody components. However, the established Lagrangian model will be quite nonlinear, whose control design usually needs the usage of the tool of feedback linearization. Another possible alternative is via coordinating two such coaxial drones for aerial manipulation tasks, in which the positions of the two coaxial drones are symmetric with respect to the geometric center of the whole combined system, in which the control design for such system can be handled based on the symmetry of this system and the control design for single coaxial drone.

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